Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/24/2005
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Project Title:	Spatial Organization within Prostate Cancer Spheroids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	End date is 9/30/2006, per P. Denkins/JSC (3/22/07)		
Key Personnel Changes/Previous PI:	O'Connor, Kim		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1351		
Performance Goal No.:			
Performance Goal Text:			
	Multicellular spheroids have a remarkable capacity to mimic solid tumors to a greater extent than cell monolayers. Spatial organization within spheroids is seminal to their in vivo-like behavior and is instrumental in the acquisition of drug resistance and expression of antigens that can serve as biomarkers for carcinogenesis. Our research provides a comprehensive analysis of the spatial profile and assembly of multicellular spheroids, using prostate cancer cells as a model system. Spheroid differentiation and viability will be determined as a function of radial position. This analysis will be performed on spheroids produced using conventional methods and in a rotating-wall vessel (RWV). The knowledge gained from this research will ultimately improve the fidelity of multicellular spheroids in the extent to which they resemble micrometastases and avascular regions of larger tumors. In future work, we will use the spheroids to develop biomarkers for carcinogenesis and countermeasure to against the harmful effects of radiation. The objective of the proposed research is to improve our understanding of how culture conditions affect the spatial profile and		

Task Description:	assembly of multicellular cancer spheroids, using prostate cancer cells as a model system. During the past year, our specific aims were to: 1) Assess the predictive capacity of our kinetic model of spheroid self-assembly. 2) Provide a detailed comparison of E-cadherin expression in prostate cancer spheroids and monolayer controls. 3) Conduct a broad investigation of the effects of spheroid formation on gene expression in well and poorly differentiated prostate cancer cells with microarray analysis. NASA Critical Path Roadmap The research described in this report is categorized in the NASA Critical Path Roadmap under the discipline of Radiation Health and addresses the risk of carcinogenesis in astronauts during the exploration of space from exposure to harmful radiation or from the synergistic effects of radiation with other space flight factors including stress. The relevant enabling question for this project is 31d: How can effective biomarkers of carcinogenic risk from space radiation be developed and validated? The overall goal of this four-year project is to produce in vitro cultures of multicellular spheroids that mimic solid tumors by gaining an understanding of tissue assembly and spheroid composition. Through this understanding, we will identify culture conditions favorable to spheroid production. In future work, we will use the spheroids to develop biomarkers for carcinogenesis and countermeasure to against the harmful effects of radiation. Background For NASA, spheroids have been the primary form of tissue reconstructed in RWV's. For the larger tissue engineering community, spheroids have a variety of applications from liver regeneration to stem-cell differentiation. When spheroids are prepared from neoplastic cells as in the proposed research, they mimic micrometastases and avascular regions of larger tumors. Spheroids have proven beneficial as a research tool to investigate cancer progression, identify biomarkers of carcinogenesis, and in the testing of new therapies and delivery strategies to combat this deadly disease. Numerous studies have demonstrated that the RWV produces multicellular spheroids that are more differentiated than cultures of the same cell line grown as a monolayer in terms of spheroid morphology and expression of proteins that are indicative of differentiated function. This is attributed to gentle mixing in the RWV that promotes mass transfer of nutrients and wastes while minimizing cellular damage from hydrodynamic forces; three-dimensional (3D) organization of cells that produces an environment in which cell shape, cell-cell and cell-matrix interactions, and interstitial fluid are akin to those in native tissue; and, for co-cultures, co-location of dissimilar cell types, mimicking the diversity found in vivo. Understanding the mechanisms of spheroid assembly will ultimately improve the fidelity of this reconstructed tissue. The kinetic models developed in this work will be useful in producing spheroids within a defined size range that have greater viability. For, neoplastic cells, model results can also aid in the analysis of cellular adhesion and the effectiveness of new therapies to inhibit tumor shedding. A critical component in the resemblance of spheroids to solid tumors is the spatial profiles within spheroids as a result of mass transport limitations and the 3D environment within the tissue. For example, the region of quiescent cells that lies beneath proliferative cells on the spheroid surface is often instrumental in the failure of tumor regression therapies. These cells have both limited accessibility and intrinsic resistance to toxic agents. Despite the importance of the spatial profile of spheroids to their function, there has been very little research done in this area with RWV's. In fact, ours will be by far the most comprehensive study of its kind for RWV spheroids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Prostate cancer is one of the most common forms of cancer in men and a leading cause of death in the United States. While surgery and radiation will successfully treat many patients with localized disease, there is still no cure for patients with advanced, metastatic prostate cancer. The proposed research will aid in the development of new cancer therapies and biomarkers for detect prostate cancer by producing tissue models that more closely approximate micrometastases than is currently possible with conventional cell-culture techniques.
	Venczel, Larson and O'Connor. Predictive capacity of kinetic model of human prostate cancer cell aggregation. Biotech Lett, submitted (2005). Attachment-dependent cells self-assemble into multicellular spheroids. When spheroids are prepared from cancer cells they exhibit a resistance to chemotherapies that approach clinical levels and have application to in vitro drug testing. Dynamics of prostate cancer cells aggregating into spheroids was modeled as a population balance to predict the spheroid size distribution in culture. The mathematical model is able to predict a priori from the culture inoculum the distribution of all spheroid size classes as a function of time with good agreement to experimental data. Using spheroids of DU 145 human prostate cancer cells as a representative system, the model was remarkable predictive of aggregation kinetics over a wide range (5-fold) of inoculum concentrations. By predicting size, it may be possible to predetermine other spheroid properties given the strong correlation between radial position within spheroids and their composition. These findings demonstrate the power of computational analysis in the design of engineered tissues. O'Connor. Designing tissues by numbers: a computational approach to tissue engineering. Biotechnol. Prog, submitted (2005). A new era in tissue engineering is emerging, one which interfaces with computer science. This trend parallels the integration of computational analysis into the biological sciences as a whole. Computation has dramatically changed the degree of complexity in research and yielded significant insight into living systems. To date milestones in tissue engineering have been achieved largely through empirical investigation. Mathematical models have the potential to significantly influence future developments in this field given the intrinsic complexity of its products. Living tissue substitutes are composed of millions of cells per cubic centimeter that assemble into 3D structures within scaffolds of native or synthetic materials. The mechanisms that signal cell growth and functional differentiation in developing tissue are highly regulated. During ex vivo engineering, cell survival requires a continuous supply of nutrients and removal of wastes in transport processes that are controlled by culture conditions within bioreactors. This review highlights representative examples by which the predictive nature of computational analysis guides experimental research to advance tissue engineering. Almagor, Barrilleaux, Clejan, David, Japa, Song, O'Connor. Comparison of E-cadherin expression in spheroid and monolayer cultures of human prostate cancer cells. Prostate, in prepartion (2005).

Task Progress:	This is a detailed analysis of the effect of spheroid formation on the expression of a single cell adhesion molecule, E-cadherin. The study was conducted over this past year, and we are preparing a manuscript on the findings for peer review. Spheroid and monolayer cultures of DU 145 and LNCaP were subjected to ELISA, RT-PCR, immunohistology and a kinetic assay. The last of these showed that (1) LNCaP cells aggregate faster into spheroids than DU 145 cells and (2) spheroids aggregate faster than single cells for both cell lines. These data are supported by the ELISA results. E-cadherin is up-regulated upon spheroid formation in both DU 145 and LNCaP cells. Of the two cell lines, LNCaP cells accumulate more E-cadherin in both spheroids and monolayer controls than DU 145 cells. The findings from RT-PCR are consistent with the ELISA data. Immunohistology demonstrates that E-cadherin expression is heterogeneous in DU 145 monolayers and homogenous in spheroids and in regions of hyperconfluence; whereas, LNCaP cells have uniform expression of the cell adhesion molecule. E-cadherin plays a seminal role in cancer progression to a metastatic state. Prior to this work, researchers assumed that E-cadherin was permanently down-regulated at the onset of metastasis. Our research shows that this is clearly not the case since both DU 145 and LNCaP were derived from metastatic tumors. Our results further suggest that spheroid formation and simulated microgravity in the HARV are useful tools to study E-cadherin expression. Influence of spheroid formation on gene expression in well and poorly differentiated prostate cancer cells (Research ongoing). To the best of our knowledge, this is the first comparison of gene expression from microarray data in multicellular spheroids and monolayer controls of cancer cells. The research for this project is still ongoing. We are subjecting spheroid and control cultures of DU 145, LNCaP and PC 3 to microarray analysis. The cultures represent a range of well (LNCaP), moderate (DU 145) and poorly (PC 3) differentiated cells. We anticipate that extent to which gene expression is up-/down-regulated by spheroid formation will be different in the three cell lines given their variation in adhesion properties (previous paragraph). Typically researchers analyze microarray data by determining the number of genes that are up-/down-regulated and by grouping genes according to their function (adhesion molecules, oncogenes, etc.). We are not only conducting the typical analysis, but are also examining how specific signaling pathways are altered by spheroid formation. In our previous studies, we demonstrated that spheroid formation alters functional differentiation of cancer cells. By examining pathways, this study will provide insight into the molecular mechanisms underlying this change in differentiation. Song, Jain, Enmon, O'Connor. Restructuring dynamics of DU 145 and LNCaP prostate cancer spheroids. In Vitro Cell Dev Biol - Anim, 40: 262-7 (2004). The research described in this publication was summarized in the 2004 NASA Annual Report. The article was listed in press in the 2004 Report and was published later in the year. Recently the article was featured in In Vitro Report. [H Song, S Jain, RM Enmon and KC O'Connor. Restructuring of prostate spheroids. In Vitro Report, Journal Highlights, 39.2 (2005)] The report features noteworthy articles from In Vitro Cell Dev Biol that have been selected by the editor.

Bibliography Type:	Description: (Last Updated: 04/03/2007)
Articles in Peer-reviewed Journals	Venczel M, Larson A, O'Connor KC. "Predictive capacity of kinetic model of human prostate cancer cell aggregation. " Biotechnol. Lett. (in press, June 2005). , Jun-2005
Articles in Peer-reviewed Journals	O'Connor KC. "Designing tissues by numbers: a computational approach to tissue engineering." Biotechnol. Prog. (submitted), December 2005. , Dec-2005
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Awards	O'Connor KC. "Who's Who Among American Teachers, May 2005." May-2005
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Presentation	O'Connor, K.; Almagor, M.; Vidulescu, C.; Clejan, S.; Song, H.; Venczel, M. "Predicting tissue assembly of prostate cancer spheroids from cell culture." European Society for Animal Cell Technology. Harrogate, North Yorkshire, UK Jun-2005
Presentation	Vidulescu, C.; Clejan, S.; O'Connor, K. "Vesicle traffic through intercellular bridges in DU 145 human prostate cancer cells." American Society for Biochemistry and Molecular Biology, San Diego, CA Apr-2005
Presentation	Vidulescu, C.; Clejan, S.; Almagor, M.; Song, H.; Venczel, M.; O'Connor, K. "Prostate cancer spheroid assembly. " NASA Cell Science Conference, Galveston, TX Feb-2005
Significant Media Coverage	Song H, Jain SK, Enmon RM, O'Connor KC. "Restructuring of prostate spheroids." In Vitro Report, Journal Highlights, 39.2, June 2005., Jun-2005